

Housing Price Prediction – Summary of Findings

This project focused on predicting house prices using machine learning techniques on the Housing Prices dataset. During the exploratory analysis, it was observed that features such as area, number of bathrooms, number of stories, air conditioning, and preferred area had the strongest influence on house prices. Larger houses with more amenities generally commanded higher prices.

Two machine learning models were developed and evaluated: Linear Regression and Random Forest Regressor. Linear Regression achieved a Mean Absolute Error (MAE) of approximately 970,043, a Root Mean Squared Error (RMSE) of approximately 1,324,507, and an R^2 score of 0.653. Random Forest Regressor achieved an MAE of approximately 1,021,546, an RMSE of approximately 1,400,566, and an R^2 score of 0.612. Based on these metrics, Linear Regression performed slightly better and was able to explain about 65% of the variation in house prices, indicating reasonably accurate predictions.

One interesting observation was that non-size-related features such as air conditioning and preferred area had a noticeable impact on housing prices. It was also surprising to see how strongly the size of the house influenced the overall price. The correlation analysis confirmed that area is one of the most important factors affecting property values.

Based on these findings, real estate businesses should focus on properties with larger areas and premium amenities, as these characteristics contribute significantly to higher prices and greater customer demand. Machine learning models can assist buyers, sellers, and agents in making data-driven decisions and improving the accuracy of property valuations.